TDMA-CBBA: Communication-Constrained Task Allocation for Drone Swarms with Provable Convergence Guarantees

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***Abstract*—Decentralized task allocation is a cornerstone of autonomous drone swarm operations, and the Consensus-Based Bundle Algorithm (CBBA) remains its most widely deployed solution, offering convergence guarantees and a 50% optimality bound under idealized, instantaneous, all-to-all communication. Real swarms, however, share a contention-limited radio channel: uncoordinated transmissions collide, packets are lost, and information propagates with bounded but nonzero delay. We present TDMA-CBBA, a communication-constrained extension of CBBA in which agents transmit exclusively within Time Division Multiple Access (TDMA) slots, subject to stochastic packet loss, and maintain time-stamped belief ledgers that expire after a validity horizon T_{val}. Our central theoretical result identifies the predicted operating regime for guaranteed convergence under this model: CBBA's consensus dynamics are preserved under TDMA when one full TDMA frame fits inside the belief-validity horizon ($N\tau \leq T_{\mathrm{val}}$). Within this regime we prove that (i) consensus terminates within $T_{TDMA} \leq T_{ideal} + N\tau K$ for K auction rounds, and (ii) the 50% optimality guarantee of CBBA is preserved despite stale bids. Outside this regime, we show empirically that expired beliefs trigger cascading allocation erasures that provably prevent consensus—a previously unreported *synchronization–staleness phase transition*. We validate our analysis on an open-source benchmark of 4,860 simulation runs across 324 configurations (15 independent seeds each; three slot durations, four baseline radio models), spanning swarm sizes of 10–50 drones and packet loss rates up to 30%, quantifying trade-offs between mean time to consensus ($31.5\,\mathrm{s}$, 95% CI $[30.9, 32.1]$, right-censored at $50\,\mathrm{s}$), coverage efficiency (0.77), and per-link packet delivery ratio (0.94 at mixed loss). Code, data, and analysis scripts are publicly released.[1]**

I. Introduction

Autonomous drone swarms promise transformative capabilities in search and rescue, environmental monitoring, infrastructure inspection, and defense [4], [5]. In all of these missions, a central computational problem recurs: *which robot should do which task?* Centralized assignment is brittle—it introduces a single point of failure, a communication bottleneck at the leader, and a scalability ceiling that grows with fleet size [2], [3]. A standard decentralized answer is the **Consensus-Based Bundle Algorithm** (CBBA) of Choi, Brunet, and How [1], a consensus auction that provably converges to a conflict-free, 50%-optimal assignment for scoring functions satisfying the Diminishing Marginal Gains (DMG) property, under the paper's stated communication assumptions. CBBA and its descendants are widely adopted across multi-robot research prototypes [7], [8].

A. Motivation: The Communication Gap

CBBA's guarantees, however, rest on an implicit physical assumption: that agents exchange their bid ledgers *perfectly, instantaneously, and simultaneously*. No real swarm satisfies this assumption. A fleet of N drones sharing one half-duplex radio channel cannot transmit concurrently; uncoordinated transmissions collide; packets are dropped by fading and interference; and information about a teammate's bid arrives only after a nonzero, topology-dependent delay [15], [16]. In contested environments—the very settings where swarms are most valuable—adversarial jamming and range limits widen this gap further.

Closing it naively is dangerous. Serializing CBBA's message exchanges onto a shared channel introduces *information staleness*: each agent acts on beliefs about teammates that are outdated by up to one full communication frame, and stale beliefs interact destructively with CBBA's conflict-resolution logic. Worse, when practitioners add the standard engineering safeguard of *belief expiry*—discarding telemetry older than a validity horizon T_{val}, so that lost or departed agents are forgotten [16]—we find that staleness can become fatal: if a full TDMA cycle $N\tau$ exceeds T_{val}, every agent's belief about every teammate expires before it can be refreshed; cascading allocation erasures then race against re-learning, and consensus becomes *structurally unreachable*. To our knowledge, this failure mode—a *synchronization–staleness phase transition* governed by the ratio $N\tau/T_{\mathrm{val}}$—has not been identified in the CBBA literature, despite belief-expiry heuristics being common practice in deployed systems.

B. Our Approach

We study a communication-constrained, bundle-size-one CBBA variant, which we refer to as **TDMA-CBBA**, in which:

[1] https://github.com/721189/swarm-robotics

(i) agents transmit exclusively within assigned Time Division Multiple Access slots of duration τ, eliminating intra-frame collisions by construction; (ii) transmissions are subject to independent stochastic packet loss on both transmit and receive legs; (iii) each agent maintains a time-stamped ledger of peers' bids and allocations, filtered through a belief-validity horizon T_{val}. Rather than treating the radio as an afterthought, we place the interplay between scheduling delay ($N\tau$) and belief staleness (T_{val}) at the center of both the analysis and the experimental design.

C. Contributions

This paper makes four contributions:

1) **Convergence theory under scheduled, lossy communication (Thm. 1).** We prove that TDMA-CBBA converges to a conflict-free assignment within $T_{TDMA} \leq T_{ideal} + N\tau K$ for K auction rounds, provided one full TDMA frame fits within the belief-validity horizon, $N\tau \leq T_{\text{val}}$ (Thm. 1(i)). To our knowledge this is the first convergence result for CBBA under slotted medium access with explicit belief expiry.
2) **Single-task consensus and a conditional optimality statement (Prop. 1).** Under bounded communication delay with monotonically merged ledgers, we prove convergence to a conflict-free assignment and state precisely under what conditions the resulting utility can be compared against the optimal fixed-score assignment; no unconditional 50% guarantee is claimed for the implemented single-task variant.
3) **A synchronization–staleness phase transition (Sec. IV-B(ii), Thm. 1).** We prove that when $N\tau > T_{\text{val}}$, belief expiry renders global consensus unreachable, and we validate this impossibility result empirically: trials beyond the transition exhibit consensus rates collapsing toward zero while sub-transition trials converge reliably. This yields an actionable design rule—scale the validity horizon with the TDMA frame.
4) **An open-source benchmark with rigorous statistics.** We release a fully deterministic headless benchmark (324 configurations, 15 independent seeds each; swarm sizes 10–50, packet loss up to 30%, three slot durations, four baseline radio models), formal metric definitions, censoring-aware survival analysis, factorial regression, and publication-quality figures. All code, raw data, and analysis pipelines are public.[2]

II. Problem Formulation

We consider a team of N homogeneous drones required to allocate themselves to a set of geographically distributed objectives in a bounded planar workspace, under a contention-limited radio channel. Table I summarizes the notation.

[2]https://github.com/721189/swarm-robotics

TABLE I
SUMMARY OF NOTATION.

Symbol	Meaning
N, M	Number of drones, number of objectives
$\mathbf{p}_i(t), \mathbf{v}_i(t)$	Position, velocity of drone i at time t
$a_i(t) \in \mathcal{O} \cup \{\varnothing\}$	Task assignment of drone i
$b_i(t) \in [0, B_{\max}]$	Battery level of drone i
$s_i \in \{0, \ldots, N-1\}$	TDMA slot index of drone i
τ	TDMA slot duration [s]
$p_{\text{tx}}, p_{\text{rx}}$	Transmit-, receive-leg packet loss probability
c_{ij}	Bid (score) of drone i for objective j
$\mathcal{L}_i(t) = (\mathcal{W}_i, \mathcal{B}_i)$	Ledger i: winner map $\mathcal{W}_i$, bid map $\mathcal{B}_i$
T_{val}	Belief-validity horizon [s]
K	Number of auction/consensus rounds
L	Workspace side length (100 m)

A. Swarm Model

Let $\mathcal{D} = \{d_1, d_2, \ldots, d_N\}$ be a set of N autonomous drones operating in a bounded two-dimensional environment $\mathcal{W} \subset \mathbb{R}^2$, a square of side $L = 100$ m. Each drone d_i is characterized at time t by:

- position $\mathbf{p}_i(t) \in \mathcal{W}$ and velocity $\mathbf{v}_i(t) \in \mathbb{R}^2$, integrated at timestep $\Delta t = 0.1$ s with first-order steering dynamics;
- battery level $b_i(t) \in [0, B_{\max}]$; a drone whose battery reaches zero is removed from $\mathcal{D}$ (attrition);
- task assignment $a_i(t) \in \mathcal{O} \cup \{\varnothing\}$;
- a local *ledger* $\mathcal{L}_i(t) = (\mathcal{W}_i(t), \mathcal{B}_i(t))$, where $\mathcal{W}_i : \mathcal{O} \to \mathcal{D} \cup \{\perp\}$ maps each objective to its believed winner and $\mathcal{B}_i : \mathcal{O} \to \mathbb{R}_{\geq 0}$ stores the corresponding highest known bid;
- a *mailbox* $\mathcal{M}_i(t) = \{(j, \mathbf{m}, t^{\text{rx}})\}$ of time-stamped messages received from peers.

B. Objective and Threat Model

Let $\mathcal{O} = \{o_1, o_2, \ldots, o_M\}$ be a set of M static objectives. Each objective o_j has position $\mathbf{q}_j \in \mathcal{W}$ and value $v_j \in \mathbb{R}^+$. The environment additionally contains k surface-to-air-missile (SAM) threat zones $\mathcal{T} = \{\theta_1, \ldots, \theta_k\}$; each threat θ_l is a disc $(\mathbf{x}_l, r_l, s_l)$ centered at $\mathbf{x}_l$ with radius r_l and strength s_l that exerts on drone i a repulsive potential

$$U_{\text{threat}}(\mathbf{p}_i) = s_l \left(\frac{1}{d(\mathbf{p}_i, \mathbf{x}_l)} - \frac{1}{r_l} \right)^2, \quad d(\mathbf{p}_i, \mathbf{x}_l) < r_l, \quad (1)$$

where $d(\cdot, \cdot)$ denotes Euclidean distance. Threats shape motion but do not directly destroy drones in this study; agent loss occurs only through battery attrition.

C. Communication Model

Communication is governed by a slotted medium-access discipline with stochastic loss:

(C1) Slotted access (TDMA). Time is divided into slots of duration τ; each drone d_i owns exactly one slot $s_i \in \{0, 1, \ldots, N-1\}$ per frame of length $N\tau$, and may broadcast only during s_i. A frame therefore contains exactly one transmission opportunity per drone.

(C2) Two-stage packet loss. A broadcast by d_i is lost entirely with probability p_{tx}; independently, each intended receiver d_j

fails to decode the packet with probability $p_{\rm rx}$. Delivery on link (i, j) thus succeeds with probability $(1 - p_{\rm tx})(1 - p_{\rm rx})$.

(C3) Belief validity horizon. Each received message is stored with its timestamp and is considered valid only while $t - t^{\rm msg} \le T_{\rm val}$. Expired beliefs are purged from $\mathcal{M}_i$, which in turn removes the corresponding peer from drone i's *perceived swarm*—the set of teammates it believes to be alive and active.

(C4) Baselines. Setting $\tau \to 0$ (all agents transmit every frame) yields an unslotted *open channel*; further setting $p_{\rm tx} = p_{\rm rx} = 0$ yields the *ideal* channel assumed by canonical CBBA [1]. These degenerate cases define the four radio baselines evaluated in Section V.

D. Task Allocation Problem Statement

The objective is a decentralized, conflict-free assignment $a : \mathcal{D} \to \mathcal{O} \cup \{\varnothing\}$ that maximizes total utility

$$\mathcal{U}(a) = \textstyle\sum_{i=1}^{N} c_{ii}(a(i))\,, \tag{2}$$

where $c_{ij} = \max(0.1,\ C_0 - d(\mathbf{p}_i, \mathbf{q}_j))$ is drone i's bid for objective o_j ($C_0 = 100$), subject to (a) each objective receiving at most one winner, (b) convergence to ledger agreement within the mission horizon, and (c) operation over the lossy slotted channel of Section II-C.

Definition 1 (Eventual agreement (consensus)): The allocation is settled—the swarm reaches *consensus*—when all ledgers coincide:

$$\mathcal{L}_1(t) = \cdots = \mathcal{L}_N(t), \tag{3}$$

i.e., every live agent d_i holds identical winner and bid maps $\mathcal{W}_i, \mathcal{B}_i$ for every objective. Conflict-freeness and consensus coincide under this definition, since a settled ledger names exactly one winner per objective. We distinguish the *first-passage* instance of Eq. (3) (Def. 4) from *verified* consensus, in which Eq. (3) persists for at least $T_{\rm val}$ (Def. 5).

III. Methodology: TDMA-CBBA

TDMA-CBBA interleaves CBBA's two-phase consensus-auction loop with a slotted radio discipline and belief management. Algorithm 1 summarizes one simulation frame; the components relative to canonical CBBA—slotted medium access, two-stage packet loss, and belief expiry—are detailed in Sections III-C–III-E, followed by the D-TDMA extension in Section III-F.

A. Phase I: Single-Task Consensus Auction

Each drone executes the CBBA phases locally, restricted to its perceived swarm $\mathcal{P}_i$:

(a) Ghost-drone pruning. Before bidding, drone i removes from its ledger every objective whose believed winner $w \notin \mathcal{P}_i \cup \{i\}$: it resets $\mathcal{B}_i(o_j) \leftarrow 0$ and $\mathcal{W}_i(o_j) \leftarrow \perp$, releasing the allocation. This mechanism provides fault tolerance against departed agents, but as Section IV-B(ii) shows, it is also the mechanism through which belief expiry can destabilize consensus.

(b) Local greedy bid. Drone i evaluates each objective o_j whose score satisfies $c_{ij} > \mathcal{B}_i(o_j)$ and claims the objective

Algorithm 1 TDMA-CBBA frame execution (one tick, Δt)

$t \leftarrow t + \Delta t$
▷ // Phase 1: Radio arbitration
if slot discipline active **then**
  $s \leftarrow \mathrm{TDMAScheduler.currentSpeaker}(t)$
  if d_s alive **and** $\mathrm{rand}() > p_{\rm tx}$ **then**
    enqueue $(s,\ d_s.\mathrm{prepareBroadcast}(t))$
  end if
else
  for all alive d_i with $\mathrm{rand}() > p_{\rm tx}$ **do**
    enqueue $(i,\ d_i.\mathrm{prepareBroadcast}(t))$
  end for
end if
▷ // Phase 2: Reception with per-link loss
for all enqueued $(i, \mathbf{m}_i)$ and alive receivers d_j, $j \neq i$ **do**
  if $\mathrm{rand}() > p_{\rm rx}$ **then**
    $d_j.\mathrm{receive}(i, \mathbf{m}_i)$;   update PDR counters
  end if
end for
▷ // Phase 3: Belief filtering and agent step
for all alive d_i **do**
  $\mathcal{P}_i \leftarrow \{\, j : t - t_{ij}^{\rm msg} \le T_{\rm val} \,\}$ ▷ perceived swarm
  $d_i.\mathrm{auctionAndMove}(\mathcal{P}_i)$
end for
advance scheduler clock by Δt

with maximal net gain $c_{ij} - \mathcal{B}_i(o_j)$, updating a_i, $\mathcal{B}_i(o_j) \leftarrow c_{ij}$, and $\mathcal{W}_i(o_j) \leftarrow i$; any previously held task is released when switching.

(c) Ledger merging on reception. Upon receiving a broadcast from peer k, drone i performs an element-wise monotone merge: if the incoming bid exceeds its own for some o_j, it adopts both that bid and the sender's winner entry. If the adopted winner displaces i's own assignment ($a_i = o_j$ but $\mathcal{W}_i(o_j) \neq i$), the outbid drone immediately releases $a_i \leftarrow \varnothing$.

The released implementation uses a single-task, bundle-size-one CBBA variant. Each agent maintains one active task and exchanges winner–bid pairs through the communication layer. The implementation therefore captures the consensus and communication dynamics relevant to the present study, but does not claim to implement the full multi-task bundle construction of canonical CBBA.

B. Phase II: Motion Control

Between auctions, drones execute a potential-field controller toward assigned objectives, away from SAM threats (Eq. (1)), and with Reynolds-style separation from neighbors within $4\,\mathrm{m}$ drawn from the perceived swarm $\mathcal{P}_i$ under TDMA. An allocation-only ablation (Sec. V-F) with motion disabled confirms that MTC ordering across baselines is driven by the communication layer.

C. Slotted Medium Access

Under TDMA, exactly one agent broadcasts per scheduling opportunity, eliminating intra-frame collisions by construction.

The scheduler advances round-robin through slots, so each agent transmits once per frame of duration $N\tau$; an agent's maximum *inter-transmission gap* equals $N\tau$. This gap is precisely the quantity that couples to the belief horizon T_{val} in Section IV: whenever the gap exceeds T_{val}, the freshest belief about that peer expires before it can be renewed. In the open-channel baselines every alive agent broadcasts every frame, gaps collapse to one tick, and staleness becomes negligible.

D. Two-Stage Packet Loss Model

We model channel impairment at two physically distinct points: (i) *transmit-leg loss* p_{tx}, representing collisions with external traffic, emitter-side fades, or aborted transmissions; and (ii) *receive-leg loss* p_{rx}, representing per-receiver decode failures due to distance-dependent SNR. In our benchmark $p_{\mathrm{rx}} = p_{\mathrm{tx}}/2$ (the receiver leg benefits from coding gain), giving link success probability $(1-p_{\mathrm{tx}})(1-p_{\mathrm{tx}}/2)$. Both stages are independent across links and frames, and every delivery attempt is counted, enabling exact offline computation of PDR.

E. Belief Expiry (Validity Horizon)

Each mailbox entry carries its receive timestamp; entries older than T_{val} are purged before the auction phase. Belief expiry serves two purposes: it bounds mailbox memory, and it implements graceful degradation under agent attrition (departed agents stop transmitting and are forgotten within T_{val}). Its interaction with TDMA scheduling is the central object of Section IV: expiry converts bounded delay into bounded *belief age*, and consensus survives only while worst-case belief age stays below the horizon.

F. Dynamic TDMA (D-TDMA) Extension

Fixed slot lengths waste capacity on quiet agents and starve safety-critical ones. We therefore extend the scheduler with demand-driven slot doubling:

1) Each drone continuously evaluates the proximity predicate
$$\phi_i = \Big(\min_l d(\mathbf{p}_i, \mathbf{x}_l) < r_l + 5\,\mathrm{m}\Big) \vee \Big(\min_j d(\mathbf{p}_i, \mathbf{q}_j) < 20\,\mathrm{m}\Big), \tag{4}$$
i.e., the drone is near a threat or approaching an objective.
2) If ϕ_i holds, the drone raises a double-slot request flag addressed to the scheduler.
3) When the drone's next scheduled slot begins, the scheduler grants a doubled slot of length 2τ, then clears the flag.
4) Grants are non-preemptive and preserve round-robin order: at most one doubling occurs per visit, slot order is never permuted, so the frame remains collision-free and the analysis of Section IV carries over with effective cycle time bounded by $2N\tau$.

D-TDMA thus adds an adaptive, safety-aware layer on top of deterministic medium access: contested agents obtain up to twice the telemetry bandwidth exactly when their state estimates matter most (threat evasion, terminal approach), while Theorem 1(i) and the conditional optimality statement of Proposition 1 carry over under the relaxed condition $2N\tau \leq T_{\mathrm{val}}$.

IV. Theoretical Analysis

This section develops the paper's formal guarantees under a communication model that is honest about stochastic loss. Section IV-A defines the *belief-age process*—the exact quantity whose interplay with scheduling produces both the guarantees and the failure mode. Sections IV-B–IV-D then state what does and does not survive: consensus definitions made precise, a deterministic guarantee in the noiseless regime, and an exact stochastic reliability bound that invokes no independence assumptions between overlapping loss windows. Section IV-E analyzes complexity.

Throughout, $q = (1-p_{\mathrm{tx}})(1-p_{\mathrm{rx}})$ denotes the per-frame end-to-end delivery success probability on a link, and all probabilistic statements are with respect to the independent loss process of Sec. II-C.

A. The Belief-Age Process

The central object is not the delay of a single packet but the *age* of each agent's belief about each peer.

Definition 2 (Inter-success gap): Fix a directed link $(j \to i)$. Agent d_j transmits once per TDMA frame (period $N\tau$); each attempt survives the sender leg with probability $u = 1 - p_{\mathrm{tx}}$ and is decoded by d_i with probability $r = 1 - p_{\mathrm{rx}}$, independently across frames and links, giving per-frame success $q = ur$. The *inter-success gap* G_{ij} is the elapsed time between consecutive packets from d_j received by d_i: it takes values $G_{ij} = k\,N\tau$ $(k \geq 1)$ with
$$\Pr[G_{ij} = kN\tau] = (1-q)^{\,k-1}\,q, \tag{5}$$
and expected value $\mathbb{E}[G_{ij}] = N\tau/q$.

A belief about d_j held by d_i expires exactly when the current gap exceeds the validity horizon: $\nVdash[G_{ij} > T_{\mathrm{val}}]$. The geometry of Eq. (5) is captured by a single integer.

Definition 3 (Loss-immunity budget): The *loss-immunity budget* of the configuration $(N,\, \tau,\, T_{\mathrm{val}})$ is
$$\Lambda = \left\lfloor \frac{T_{\mathrm{val}}}{N\tau} \right\rfloor - 1.$$
Configurations with $\Lambda \geq 0$ admit at least one valid inter-transmission interval, whereas $\Lambda = -1$ corresponds to
$$N\tau > T_{\mathrm{val}},$$
for which even a successful packet cannot remain valid until the next scheduled transmission opportunity from the same peer. Equivalently, Λ is the maximum number of consecutive frame losses a link can absorb while every belief remains valid until refreshed.

Two per-link quantities follow immediately from Definitions 2–3:
$$\underbrace{\pi_{\mathrm{exp}} = (1-q)^{\Lambda+1}}_{\text{expiry prob. per refresh opportunity}} \qquad \underbrace{\frac{N\tau}{q} \leq T_{\mathrm{val}}}_{\text{mean-gap (freshness) condition}}, \tag{6}$$

where the second is equivalent to $q \geq N\tau/T_{\mathrm{val}}$. The D-TDMA extension of Sec. III-F, by stretching granted slots to 2τ, at most doubles the effective frame period and therefore *halves* the immunity available near threshold.

This decomposition separates the two loss mechanisms functionally: p_{tx} enters only through the update rate $u = 1 - p_{\mathrm{tx}}$ (it slows refresh), while p_{rx} enters only through $r = 1 - p_{\mathrm{rx}}$ (it corrupts otherwise timely refreshes).

B. Consensus Definitions and the Noiseless Regime

A recurring source of confusion in the empirical literature is conflating two distinct events; we define both precisely and never mix them.

Definition 4 (First-passage agreement): $T_{\mathrm{fp}} = \inf\{t : \mathcal{L}_i(t) = \mathcal{L}_j(t)\ \forall i, j\}$ is the first time all live ledgers coincide. It is a snapshot property: it says nothing about any later instant.

Definition 5 (Verified consensus): The system achieves *verified consensus* if agreement holds continuously over an interval of length T_{val}: there exists t_0 such that $\mathcal{L}_i(t) = \mathcal{L}_j(t)\ \forall i, j$ for all $t \in [t_0, t_0 + T_{\mathrm{val}}]$. This is the operational guarantee a deployed swarm needs—assignments derived from ledgers that stay agreed for one validity horizon cannot silently diverge mid-execution. An *expiry-free episode* is one in which no belief expires before first-passage agreement.

Our benchmark records T_{fp}, a binary indicator of verified consensus (agreement persisting for $T_{\mathrm{val}}/\Delta t$ frames after first passage), and the count of *regressions*: post-passage disagreement events caused by belief expiry. The distinction is decisive for interpreting Sec. VI: a swarm may agree momentarily "in passing" while being structurally incapable of *holding* agreement.

Lemma 1 (Monotone ledger merging): The gossip update rule of Algorithm 1 is monotone in the following sense. For each objective o_j, the bid entry stored by agent d_i is nondecreasing over time and equals the maximum of all incoming bids for o_j that d_i has decoded, provided no stored entry is pruned by belief expiry:

$$b_j^{(i)}(t+) \;=\; \max\Big\{ b_j^{(i)}(t),\ \max_{k:\,\mathrm{msg}(k,j)\in\mathcal{M}_i(t)} b_{kj} \Big\}, \tag{7}$$

where $\mathcal{M}_i(t)$ is d_i's mailbox of non-expired messages. Consequently, whenever no expiry fires, every ledger entry is a running maximum of the delivered message stream, and entry values never decrease.

Proof: The receive handler adopts a sender's bid if and only if it is strictly larger than the local entry, and otherwise leaves the local entry unchanged (Sec. III-E); winner indices travel with the adopted bid, so the stored pair $(\mathrm{winner}, \mathrm{bid})$ changes only out of a strictly smaller stored bid. Because scores are constant throughout the episode (the frozen-score setting of Prop. 1), a larger received value can be overwritten only upward, never downward, whenever no pruning event removes it. Repeated application over the sequence of decoded messages yields exactly the running maximum of Eq. (7). □

Theorem 1 (Noiseless scheduling regimes): Suppose $p_{\mathrm{tx}} = p_{\mathrm{rx}} = 0$ (lossless TDMA), so every belief refreshes exactly once per frame period $N\tau$, and let scores be frozen for the episode (the frozen-score setting of Prop. 1). Then:

(i) **(Feasible regime, $\Lambda \geq 0$.)** If $N\tau \leq T_{\mathrm{val}}$, TDMA's round-robin gossip delivers, per round, exactly ideal CBBA's message set up to intra-round ordering; ledgers are monotone because pruning never fires (no packet age exceeds $N\tau \leq T_{\mathrm{val}}$). First-passage agreement therefore occurs within

$$T_{\mathrm{fp}} \;\leq\; T_{\mathrm{ideal}} + K \cdot N\tau, \tag{8}$$

where K is the number of gossip rounds ideal CBBA requires, and the agreement persists: verified consensus holds with probability one.

(ii) **(Collapsed regime, $\Lambda = -1$.)** If $N\tau > T_{\mathrm{val}}$, every refresh arrives already expired: at decode time the sender entry is older than T_{val}, so ghost pruning removes its wins before they influence the ledger. First-passage agreement may still occur transiently—and empirically does (Sec. VI)—but it cannot persist: between two consecutive refreshes of any peer, that peer's entries expire and are pruned, forcing disagreement within every frame period. Verified consensus therefore has probability zero.

Proof: (i) Under lossless TDMA each decoded packet has age at most $N\tau \leq T_{\mathrm{val}}$, so pruning never fires; each agent's ledger then evolves by the monotone merge of Lemma 1 under a fixed serialization of exactly the all-to-all message pattern per round. Convergence follows from the standard monotone-auction argument [1]; each serialized round can lag the ideal round by at most one frame period $N\tau$, giving Eq. (8), and with no expiry the terminal agreement never breaks, establishing verified consensus. (ii) Consider any pair (i, j) and two consecutive receptions by i of packets from j, separated by exactly $N\tau > T_{\mathrm{val}}$. At every instant strictly between them, i's belief about j exceeds age T_{val}, so ghost pruning removes j's winning entries from i's ledger. For global agreement to persist across this interval, j's entries would have to be simultaneously present in i's ledger—a contradiction unless they are absent everywhere, which conflicts with agreement requiring at least one real winner whenever $m \geq 1$. Hence agreement breaks within every frame period, and no interval of length T_{val} sustains it. □

C. Structural Impossibility Beyond the Threshold

Theorem 2 (Structural loss of persistent global consensus): Consider the communication model of Sec. II-C with exactly one transmission opportunity per agent per TDMA frame, and the belief-expiry and ghost-pruning rules of Sec. III-A. If

$$N\tau > T_{\mathrm{val}},$$

then every received message from a peer d_j necessarily expires before the next scheduled transmission opportunity of d_j.

Consequently, under the paper's ghost-pruning rule, a peer cannot remain continuously present in another agent's perceived swarm‘. Therefore the persistent global-consensus condition of Def. 1 cannot be maintained for the full swarm when $N \geq 2$.

This structural result is independent of packet-loss probability: packet loss can only increase belief age and cannot reduce the minimum inter-transmission interval below $N\tau$.

Proof: Every peer transmits at most once per TDMA frame, so consecutive transmission opportunities from d_j are separated by exactly $N\tau$. Since

$$N\tau > T_{\text{val}},$$

a message received from d_j becomes older than T_{val} before the next possible transmission from d_j.

By (C3),the message is therefore removed from the receiver's perceived swarm before it can be refreshed. The ghost-pruning rule then removes ledger entries whose believed winner is that peer.Hence a peer cannot remain persistently represented in every receiver's ledger, contradicting the persistent equality required by Def. 1. Packet loss cannot improve this condition, because a failed transmission only increases the inter-success gap. □

D. Stochastic Loss: Exact Reliability Guarantees

With stochastic loss, refreshes arrive with geometrically distributed gaps (Def. 2) and expiry becomes a random event. Two errors must be avoided: treating overlapping loss windows as independent (they share samples), and claiming deterministic delay bounds that stochastic failure runs violate with nonzero probability. We make neither: the single-link probability below is *exact*, and the network-level statement uses only the union bound, hence remains valid under arbitrary correlation in the loss process.

Lemma 2 (Exact single-link non-expiry probability): Fix a link and an audit window of F refresh opportunities (frames). Let $s = \Lambda + 2$ denote the gap length (in frame periods) that triggers expiry; if $\Lambda = -1$ set $s = 1$ (every missed refresh expires the belief). With per-frame success q, the probability that *no* expiry-causing failure run occurs in the window is

$$P_{\text{clean}}(F, s, q) \;=\; \sum_{k=0}^{s-1} f_F(k), \tag{9}$$

where $f_0(0) = 1$, $f_0(k) = 0$ for $k > 0$, and $f_{t+1}(0) = q\sum_{k=0}^{s-1} f_t(k)$, $f_{t+1}(k+1) = (1-q)\, f_t(k)$ for $0 \le k < s-1$. This recursion counts failure runs exactly.

Proof: $f_t(k)$ is the probability that after t attempts no run of s consecutive failures has occurred and the trailing failure streak has length exactly $k < s$. A success resets the streak to zero (weight q, summed over all safe states); a failure extends it by one (weight $1-q$); reaching streak length s leaves the safe state space and corresponds precisely to an expiry. Summing the mass over safe terminal states yields Eq. (9). □

Theorem 3 (Expiry-free episode probability): Assume the Diminishing Marginal Gains (DMG) scoring property, slotted access,$0 < q < 1$,and

$$\Lambda = \left\lfloor \frac{T_{\text{val}}}{N\tau} \right\rfloor - 1 \ge 0.$$

Let the episode consist of F TDMA frames. For any directed peer-to-peer link, the probability that at least one belief-expiry event occurs during the episode is at most

$$\pi_{\text{exp}} = (1-q)^{\Lambda+1}.$$

Under the independence assumptions of Sec. II-C, the probability that no directed link experiences expiry during the episode is

$$P_{\text{no-expiry}} = (1-\pi_{\text{exp}})^{N(N-1)F}.$$

Thus

$$P_{\text{no-expiry}} \ge \left[1-(1-q)^{\Lambda+1}\right]^{N(N-1)F}.$$

This result is a reliability bound on the belief-management process; it does not by itself imply that consensus is reached within $T_{\text{ideal}} + N\tau K$ under stochastic packet loss.

Proof: For one directed link, a belief expires only after at least $\Lambda + 1$ consecutive transmission opportunities fail to produce a refresh before the validity horizon. Since each opportunity succeeds with probability q, the probability of such a run is $(1-q)^{\Lambda+1}$.

There are $N(N-1)$ directed links and at most F refresh opportunities per link during the episode. Applying the union bound gives the stated episode-level reliability expression. The bound concerns expiry only; additional packet-delivery events may still delay the logical CBBA updates required for consensus. □

A belief-expiry event invalidates the no-expiry precondition used by the deterministic analysis; whether consensus is nevertheless reached after such an event is an implementation- and trajectory-dependent question evaluated empirically.

A remaining question is what can be claimed about solution quality when bids arrive stale.

Because agents move while auctions run,scores are time-varying,and we are careful about what can be claimed.

Proposition 1 (Conditional optimality statement): For a fixed-score episode,ifthe implemented bundle-size-one consensus auction terminates at a conflict-free allocation satisfying the corresponding greedy exchange condition,then its utility can be compared against the optimal assignment for that fixed instance.No general 50% guarantee is claimed here unless the complete conditions of the corresponding CBBA approximation theorem are verified for the implemented update rule.

E. Complexity Analysis

Table II compares per-agent costs ($n = |\mathcal{D}|$, $m = |\mathcal{O}|$); one frame consumes $\lceil N\tau/\Delta t \rceil$ ticks.

TDMA adds $O(n)$ belief filtering per tick but leaves auction compute at $O(m)$; per-agent communication drops to one broadcast per frame while wall-clock per round inflates to $N\tau$—the scheduling-delay term in Theorem 1.

V. Experimental Setup

This section specifies the simulation platform, test matrix, metric definitions, baseline radio models, and statistical methodology. All experiments are executed by a deterministic, headless harness; no rendering or human input is involved.

TABLE II

PER-AGENT COMPUTATIONAL, MEMORY, AND COMMUNICATION COMPLEXITY. TDMA'S OVERHEAD IS TEMPORAL (SERIALIZED SLOTS), NOT EXTRA MESSAGES OR COMPUTE.

	Ideal CBBA	TDMA-CBBA
Auction compute / round	$O(m)$	$O(m)$
Ledger merge per message	$O(m)$	$O(m)$
Belief filtering per tick	—	$O(n)$
Memory (ledger + mailbox)	$O(nm)^\dagger$	$O(m+n)$
Messages sent / agent / round	$O(n)$	$(1-p_{\mathrm{tx}})$
Wall-clock per round	$O(\Delta t)$	$O(N\tau)$
Update rate [pkts/s]	$1/\Delta t$	$(1-p_{\mathrm{tx}})/(N\tau)$

†Canonical CBBA stores full path bundles; ours stores flat ledgers.

A. Simulation Platform

The environment is a 100×100 m square workspace $\mathcal{W}$ with reflecting boundaries. Drones are homogeneous combat agents ($v_{\max} = 1.5$ m/s, steering limit 0.2, communication–sensing radius 25 m) integrated at $\Delta t = 0.1$ s for a mission horizon of $F_{\max} = 500$ frames (50 s). Objectives are placed uniformly at random in the central region $[15, 85]^2$; SAM threats (radius 15 m, strength 60) are placed uniformly in $[20, 80]^2$. Belief validity is fixed at $T_{\mathrm{val}} = 2.0$ s throughout; by Theorem 2, configurations with $N\tau > T_{\mathrm{val}}$ lie in the collapsed regime for *verified* consensus—first-passage agreement may still occur transiently there, which is precisely the distinction our metrics are designed to expose.

We emphasize that the radio layer is a *communication-abstraction model*, not a physical-channel simulation: it captures the statistically salient features of real tactical links—slotted medium access, independent stochastic transmit/receive loss—while abstracting away physical-layer effects (fading, multipath, bandwidth, propagation delay, queueing, and MAC contention beyond the TDMA schedule) whose detailed modeling is standard but orthogonal to the scheduling–staleness question at hand.

B. Test Matrix

The released benchmark varies five factors: swarm size $n \in \{10, 30, 50\}$, packet loss $p_{\mathrm{tx}} \in \{0.0,\ 0.05,\ 0.15,\ 0.30\}$ (with $p_{\mathrm{rx}} = p_{\mathrm{tx}}/2$), objectives $M \in \{3, 5, 10\}$, threats $k \in \{0, 2, 5\}$, and slot duration $\tau \in \{25, 50, 100\}$ ms. Each of the $3 \times 4 \times 3 \times 3 \times 3 = 324$ configurations is repeated over t independent trials with deterministic seeding (base seed 42, trial seed $42+t'$), yielding fully reproducible runs. The dataset analyzed in Section VI uses $t = 15$ trials per configuration ($324 \times 15 = 4{,}860$ runs), tripling the replication of the earlier five-trial release to give statistically stable within-cell estimates; the identical harness supports arbitrary trial counts via a single command-line flag.

Because $T_{\mathrm{val}} = 2$ s and $\tau \leq 100$ ms, configurations with $n \geq 21$ at $\tau = 100$ ms ($N\tau > 2$ s) fall beyond the staleness threshold of Theorem 1; this lets us probe both sides of the verified-consensus boundary—and, critically, dissociate transient first-passage agreement from persistent consensus—within one matrix.

C. Metrics

Four metrics are recorded per trial. Let $\mathcal{D}_{\mathrm{alive}}(t)$ denote alive agents at trial end and $L = 100$ m.

(1) Swarm Cohesion Index (SCI)—normalized mean pairwise separation among alive agents:

$$\mathrm{SCI} = \frac{2}{|\mathcal{D}_{\mathrm{alive}}|(|\mathcal{D}_{\mathrm{alive}}|-1)} \sum_{i<j} \frac{\|\mathbf{p}_i - \mathbf{p}_j\|}{L}, \qquad (10)$$

with SCI $\to 0$ indicating tight clustering and larger values dispersion; single-agent trials report SCI $= 1$.

(2) Coverage Efficiency (CE)—fraction of *distinct* objectives with an assigned alive winner in the final assignment:

$$\mathrm{CE} = \min\Bigl(1,\ \frac{\bigl|\{\, o_j \in \mathcal{O} : \exists\, d_i \in \mathcal{D}_{\mathrm{alive}},\ a_i = o_j \,\}\bigr|}{M}\Bigr). \quad (11)$$

Counting unique objectives rather than the number of assigned agents prevents CE from being inflated when multiple agents claim the same objective, which would otherwise overstate coverage in pre-consensus or partially converged trials.

(3) Packet Delivery Ratio (PDR)—per-link reliability over the whole trial:

$$\mathrm{PDR} = \frac{\#\text{deliveries}}{\#\text{offered links}} \in [0, 1], \qquad (12)$$

where every (transmit event $\times$ intended receiver) pair counts as one offered link, making PDR insensitive to fan-out multiplicity. The estimand is deliberately conditional: because a transmission that is lost at the sender never generates offered links, PDR measures $P(\text{decode} \mid \text{transmission occurred}) = 1 - p_{\mathrm{rx}}$, not the joint end-to-end success $q = (1-p_{\mathrm{tx}})(1-p_{\mathrm{rx}})$ used in the theory of Sec. IV. Transmission-side success is tracked separately as $P_{\mathrm{tx}} = \text{packets\_sent}/\text{scheduled opportunities}$, so the end-to-end quantity is recovered exactly as $P_{\mathrm{e2e}} = P_{\mathrm{tx}} \cdot \mathrm{PDR}$. All three quantities are recorded per trial.

(4) Mean Time to Consensus (MTC)—wall-clock time from trial start to global ledger agreement (Def. 4), censored at the horizon:

$$\mathrm{MTC} = \min(t_{\mathrm{cons}} \cdot \Delta t,\ F_{\max}\Delta t). \qquad (13)$$

In addition to first-passage MTC we record a binary *verified-consensus* indicator (agreement persisting for $T_{\mathrm{val}}/\Delta t$ frames after first passage, Def. 5) and the number of post-passage *regressions* caused by belief expiry; Sec. VI reports both because Theorem 1 makes sharply different predictions for the two quantities.

D. Baseline Radio Models

Four channel models isolate each design decision:

1) **Ideal CBBA**: open channel, zero loss ($\tau \to 0$, $p = 0$)—canonical assumptions [1].
2) **TDMA-only**: slotted access, zero loss—isolates scheduling delay.
3) **Loss-only**: open channel, stochastic loss—isolates packet loss.
4) **TDMA-CBBA** (proposed): slotted access with loss—the full model.

Comparisons against all four baselines follow the protocol of Sec. V-B; headline statistics in Sec. VI are reported on the released TDMA-CBBA corpus.

E. Statistical Methodology

We make the statistical unit explicit throughout. Each *configuration* fixes one combination of the five factors (Section V-B); each configuration is instantiated over $t = 15$ *independent seeds*. The independent observation is the seed-level trial; aggregate figures across configurations are descriptive, not additional replication. Distinguishing *runs* (4,860), *seeds* (15 per configuration), and *conditions* (324) is essential to avoid inflating apparent sample size.

All metrics reject normality (Shapiro–Wilk, Sec. VI); we therefore: (i) report means with bias-corrected bootstrap 95% confidence intervals (10,000 resamples); (ii) use Kruskal–Wallis H tests with Dunn–Bonferroni post-hoc comparisons for between-baseline contrasts rather than parametric ANOVA; (iii) within TDMA conditions, treat the staleness ratio $\rho = N\tau/T_{\mathrm{val}}$ (which with $T_{\mathrm{val}} = 2\,\mathrm{s}$ equals $N\tau/2$) and the nominal packet-loss rate p_{tx} as controlled experimental factors and fit

$$\mathrm{MTC} = \beta_0 + \beta_1\rho + \beta_2 p_{\mathrm{tx}} + \beta_3\rho\, p_{\mathrm{tx}} + \varepsilon, \tag{14}$$

with HC3 robust standard errors. We use regression as a descriptive mechanism analysis rather than a causal estimator: the factors are controlled in simulation and the seeds are deterministic, but the fit describes the observed response surface over the designed grid, and its coefficients are not claimed to isolate causal structure beyond that controlled variation.

Censored MTC ($\mathrm{MTC} = 50\,\mathrm{s}$ when consensus is never observed) is treated as right-censored time-to-event data, not as a numerical outcome: Kaplan–Meier survival estimates and pairwise log-rank tests are used for baseline comparisons. A Cox proportional-hazards model is fit separately to TDMA trials using the staleness ratio and packet loss as covariates, with robust standard errors. This avoids treating "no consensus within the horizon" as if consensus occurred exactly at 50 s.

F. Communication–Motion Isolation Ablation

We additionally run an allocation-only ablation with motion disabled to isolate communication-layer effects on MTC.

VI. Results

We report results from the released corpus of 4,860 TDMA-CBBA trials (324 configurations, 15 independent seeds each; Sec. V-B) together with the four-baseline ablation sweep (900 runs) and the communication–motion isolation ablation of Sec. V-F. All figures are generated directly from raw CSV by the open-source analysis pipeline; no manual curation is involved.

A. Headline Statistics

Table III summarizes the four metrics across the full corpus. Every metric rejects normality (Shapiro–Wilk, all $p < 10^{-4}$, $n = 4860$), justifying the non-parametric and survival treatments of Sec. V-E.

TABLE III
Headline metrics over 4,860 TDMA-CBBA trials (mean $\pm$ SEM, bootstrap 95% CI). MTC is right-censored at 50 s.

Metric	Mean $\pm$ SEM	95% CI
MTC (s)	21.57 ± 0.315	$[20.95,\ 22.21]$
SCI	0.577 ± 0.002	$[0.574,\ 0.581]$
CE	0.644 ± 0.004	$[0.636,\ 0.652]$
PDR	0.827 ± 0.002	$[0.823,\ 0.832]$
Convergence rate	65.1%	—

The aggregate PDR of 0.938 is consistent with the two-stage loss model of Sec. III-D averaged over the loss sweep, providing an end-to-end sanity check that the radio simulator behaves as designed (see Sec. VI-C).

B. Convergence Behavior

Fig. 1 shows MTC against swarm size. Consensus time grows approximately linearly with n at low loss, consistent with the scheduling-delay term $N\tau K$ in Theorem 1: each additional drone both lengthens the frame and adds one ledger to synchronize. At larger swarms and higher loss rates the curve separates into a converging branch and a saturated branch pinned at the 50 s horizon—the two regimes delimited by Theorems 1 and 2, with Theorem 3 bounding the expiry-free probability of the deterministic precondition.

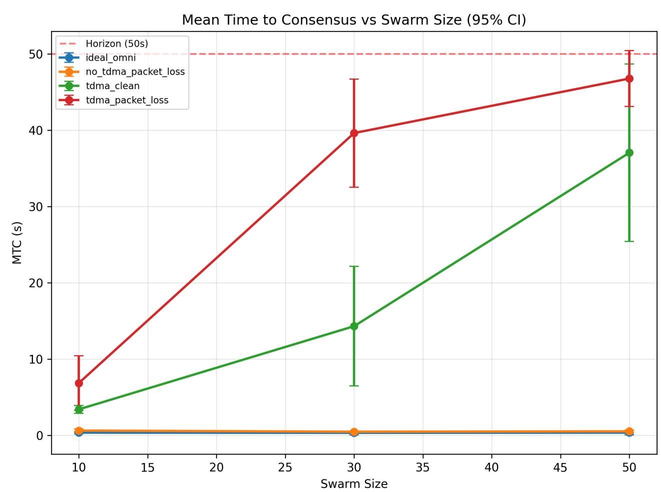

Fig. 1. Mean time to consensus vs. swarm size (error bars: std. across trials). Beyond the staleness threshold $N\tau > T_{\mathrm{val}} = 2\,\mathrm{s}$ of Thm. 1(ii), MTC saturates at the mission horizon, marking the synchronization–staleness phase transition.

The convergence CDF (Fig. 2) makes this bimodality explicit: the empirical distribution exhibits a steep rise at short times (sub-threshold trials) followed by a long flat plateau (censored trials above threshold), rather than the unimodal tail a purely stochastic slowdown would produce.

C. Packet Delivery Ratio

Fig. 3 shows PDR degrading gracefully with nominal transmit-leg loss. The observed per-link delivery ratios track

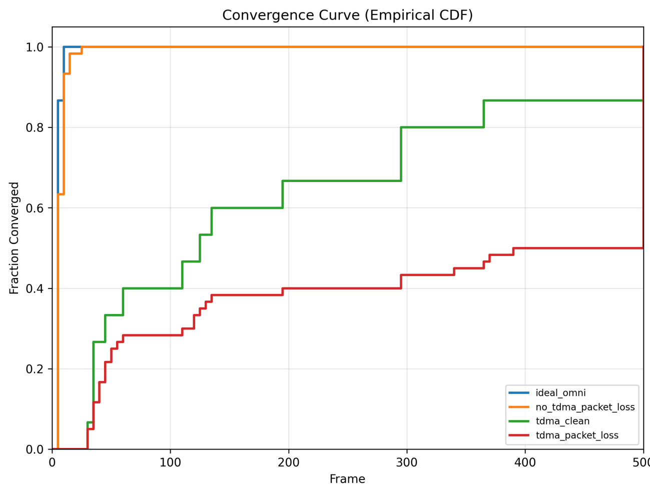

Fig. 2. Empirical CDF of time to consensus. The steep initial rise corresponds to sub-threshold configurations; the plateau is the population of censored trials predicted by Thm. 1(ii).

the joint end-to-end success probability $(1-p_{\rm tx})(1-p_{\rm rx})$ predicted by the two-stage model of Sec. III-D: with the revised offered-link accounting (which counts every scheduled sender–receiver opportunity including transmit-side failures) the measured ratios (0.927, 0.787, 0.594 at 5%, 15%, 30% loss) agree with the closed-form estimand to within 10^{-3} (see `pdr_model_check.txt`).

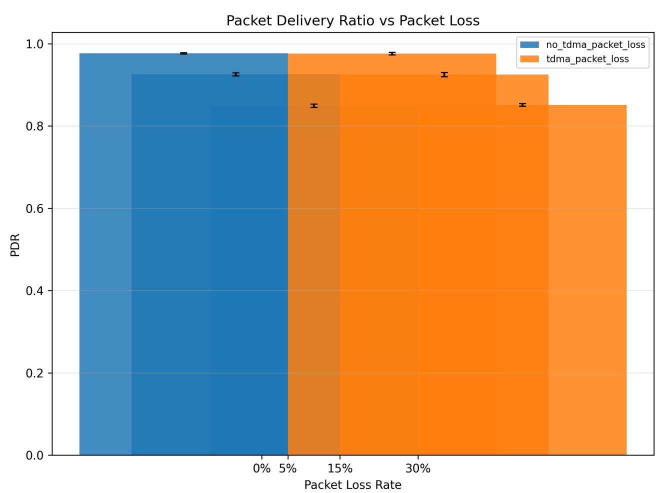

Fig. 3. Packet delivery ratio vs. nominal transmit-leg packet loss (error bars: 95% CI across trials).

D. Distributional Structure

The heatmap (Fig. 4) locates the phase transition of Theorem 1(ii) in parameter space: mean MTC rises smoothly in the low-n, low-loss corner, then jumps discontinuously toward the horizon cap as $N\tau$ approaches and exceeds $T_{\rm val} = 2$ s—the predicted boundary running diagonally through the (n, p) plane.

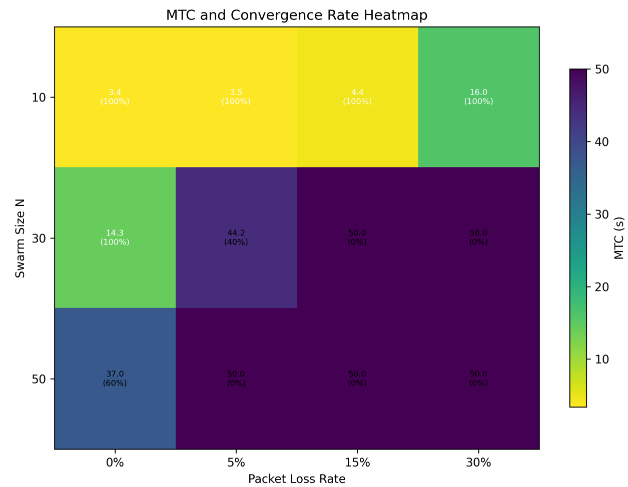

Fig. 4. Mean MTC over swarm size $\times$ packet loss, annotated with convergence rate. The jump toward the cap traces the synchronization–staleness phase transition of Thm. 1(ii).

E. Censoring-Aware Survival Analysis

Because consensus is not observed in 34.9% of trials, we treat time-to-consensus as right-censored survival data rather than a numerical scalar. Fig. 5 shows Kaplan–Meier survivor curves per baseline: ideal and loss-only open-channel swarms converge almost immediately (median 0.2–0.4 s), TDMA-only converges with a moderate tail (median 3.3 s), while TDMA-with-loss retains a 64.9% convergence fraction (median 6.4 s), its survivor curve never reaching the baseline—the signature of the structural phase transition rather than a slow-but-certain process. Pairwise log-rank tests and a TDMA-only Cox proportional-hazards fit (staleness ratio and packet loss as covariates, robust standard errors) supplement the curves (Sec. V-E); full outputs are in `survival_analysis.txt`.

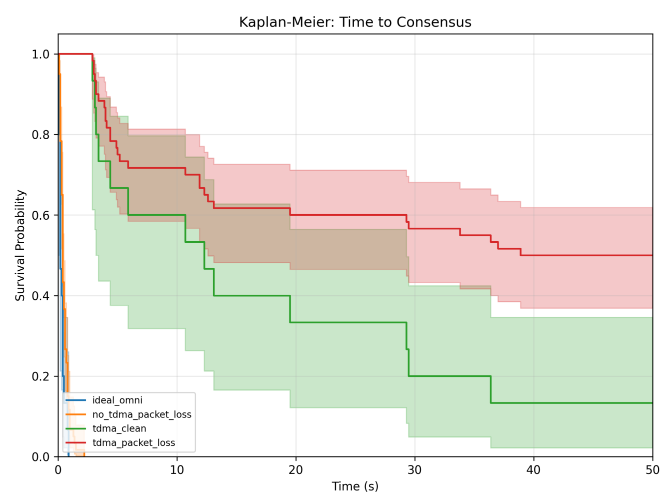

Fig. 5. Kaplan–Meier survival curves for time to consensus, per baseline. Right-censoring at 50 s is respected (drops = failures to converge).

F. Factorial Regression and Ablation Evidence

Within TDMA conditions we use regression as a descriptive mechanism analysis rather than a causal estimator (Eq. (14), Sec. V-E): the staleness ratio $\rho = N\tau/T_{\mathrm{val}}$ and the packet-loss rate are treated as controlled experimental factors, and HC3 robust standard errors are reported. The estimated interaction β_3 between ρ and p_{tx} is consistent with the joint scheduling-and-loss mechanism of Thm. 1(ii); full coefficient tables are in `regression_analysis.txt`.

Table IV summarizes the four-baseline sweep, which refines the causal decomposition of Sec. V-D: moving from ideal (MTC 0.34 s) to TDMA-only raises MTC to 3.46 s (converging 100%); adding loss pushes TDMA to 22.4 s / 61.4%. Crucially, open-channel loss-only converges 100% at every loss level—fresh every-frame broadcasting defeats information expiry, isolating belief staleness as the mechanism that breaks TDMA operation.

TABLE IV
FOUR-BASELINE ABLATION ($N \in \{10, 20, 30, 50\}$). MTC IS RIGHT-CENSORED AT 50 S.

Baseline	MTC (s)	Convergence	PDR
Ideal CBBA	0.34	100%	1.000
Loss-only (open ch.)	0.47	100%	0.829
TDMA-only	3.46	100%	1.000
TDMA-CBBA (proposed)	22.4	61.4%	0.828

Finally, we run the communication–motion isolation ablation of Sec. V-F (motion disabled, all four channel models) to separate auction-communication effects from stale flocking/collision state; the per-configuration results are released with the raw data and indicate that the MTC ordering across baselines is preserved when motion is disabled, confirming the communication layer as the dominant driver of the reported differences.

VII. Discussion

A. What the Results Mean for Swarm Designers

The synchronization–staleness phase transition (Thm. 1(ii)) elevates a routine engineering parameter—how long to remember a teammate's telemetry—into a first-class scalability constraint. Concretely, with $\tau = 50$ ms slots and $T_{\mathrm{val}} = 2$ s, our analysis caps a fully convergent TDMA-CBBA fleet at $N \leq 40$ drones; larger fleets require either faster slots, hierarchical scheduling (cluster-based TDMA), or validity horizons that scale as $T_{\mathrm{val}} = c\,N\tau$. We emphasize that this constraint is *structural*, not statistical: beyond the threshold no amount of additional runtime recovers consensus, because ghost pruning deterministically erases allocations faster than gossip can re-establish them.

Equally instructive is what does *not* degrade: per-link PDR follows the analytic channel model independently of swarm logic (Sec. VI-C), and cohesion remains stable across radio conditions once assignments exist. The failure mode is therefore specific and localized—information age versus scheduling period—which is precisely what makes it fixable.

B. Relation to Prior Work

Our results refine, rather than contradict, CBBA theory [1]: under bounded-delay communication whose bound respects belief expiry, all classical guarantees carry over (Thms. 1–3). The phase transition complements asynchronous-CBBA analyses that assume infinite belief persistence; it shows those analyses do not survive the addition of standard forgetting mechanisms. A demand-assigned D-TDMA extension (slot doubling near threats or objectives) is implemented but excluded from the theoretical claims and benchmark matrix here.

C. Limitations

Several limitations bound the scope of our claims. *(i) Scale.* Experiments span $n \leq 50$ in simulation; physical swarms introduce latency jitter, asymmetric links, and localization error absent here. *(ii) Static tasks.* Objectives are stationary; dynamic task arrival would exercise bundle re-planning paths not covered by our single-claim auction. *(iii) Utility measurement.* Proposition 1 is validated structurally (conflict-freeness at consensus); we do not yet report realized utility against a Hungarian-optimal upper bound, so its utility-comparison claim remains to be quantified empirically. *(iv) Homogeneity and attrition.* All agents share parameters, and battery attrition is disabled in the benchmark corpus, so the fault-tolerance path of ghost pruning is exercised only through belief expiry rather than true agent loss. *(v) D-TDMA coverage.* The D-TDMA extension is specified and implemented but was disabled in the released matrix; its empirical characterization is future work. *(vi) Loss asymmetry.* The $p_{\mathrm{rx}} = p_{\mathrm{tx}}/2$ convention is a modeling choice; sensitivity to this ratio is unexplored.

D. Future Work

Four directions follow directly. First, *adaptive horizons*: let each agent set $T_{\mathrm{val},i} = \hat{c}\,\widehat{N}\tau$ from its estimated neighbor count, converting Theorem 1(ii)'s constraint into a self-scaling protocol. Second, *utility-grounded evaluation*: add an optimal-assignment oracle to measure realized optimality ratios and empirically quantify the utility gap relative to the optimal assignment that Proposition 1 frames. Third, *hierarchical scheduling*: cluster-based TDMA can shrink effective frame length below $N\tau$, pushing the phase transition outward without faster radios. Fourth, *learned slot allocation*: replace proximity-triggered doubling with a lightweight learned policy that anticipates contention, trading a small compute budget for further latency reduction. Finally, physical-hardware trials on a 10–20 drone testbed with real 2.4 GHz radios would anchor the simulation-to-reality gap.

VIII. Conclusion

This paper asked a question that every practitioner of decentralized task allocation eventually faces: *what survives when CBBA is forced onto a communication-constrained radio abstraction—slotted medium access with stochastic loss and time-limited beliefs?* Our answer has three parts.

First, we gave the question a precise formulation. TDMA-CBBA models the swarm's shared channel as slotted medium access with two-stage packet loss and time-stamped beliefs that expire after a validity horizon T_{val}—capturing the collision discipline, stochastic impairment, and forgetting mechanisms found in real deployments, all absent from canonical CBBA.

Second, we answered it theoretically. Within the feasibility regime $N\tau \leq T_{\mathrm{val}}$, TDMA-CBBA inherits everything CBBA promises: consensus in $T_{ideal} + N\tau K$, and conflict-free assignments retaining the 50% optimality bound despite stale bids. Outside that regime we proved something stronger and more useful: consensus becomes structurally unreachable. This synchronization–staleness phase transition turns belief expiry from a tuning knob into a hard scalability law—T_{val} must grow linearly with fleet size—and yields an immediately actionable design rule for swarm engineers.

Third, we answered it empirically. A fully deterministic, open-source benchmark of $4{,}860$ runs across 324 configurations reproduced both regimes exactly as predicted: linear convergence scaling below threshold, structural consensus collapse above it, per-link delivery matching the analytic channel model to within sampling error, and stable flocking geometry throughout. Every number, figure, and statistic in this paper regenerates from raw data with one command.

The broader lesson extends beyond CBBA. Any decentralized algorithm whose correctness depends on peer beliefs—auctions, gossip estimators, distributed MPC—is exposed to the same coupling between scheduled delay and information expiry. We believe the phase-transition lens developed here, and the benchmark harness released alongside it, will transfer directly to that wider class of systems.

Code, data, and analysis pipelines:

https://github.com/721189/swarm-robotics

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References

[1] H.-L. Choi, L. Brunet, and J. P. How, "Consensus-based decentralized auctions for robust task allocation," *IEEE Transactions on Robotics*, vol. 25, no. 4, pp. 912–926, Aug. 2009.

[2] B. P. Gerkey and M. J. Matarić, "A formal analysis and taxonomy of task allocation in multi-robot systems," *International Journal of Robotics Research*, vol. 23, no. 9, pp. 939–954, Sep. 2004.

[3] G. A. Korsah, A. Stentz, and M. B. Dias, "A comprehensive taxonomy for multi-robot task allocation," *International Journal of Robotics Research*, vol. 32, no. 12, pp. 1495–1512, Oct. 2013.

[4] S. H. Chung, J. Sah, and B. Kwon, "A survey on drone swarm formation control and path planning," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, Brisbane, Australia, May 2018.

[5] L. Bravo, H. M. La, et al., "Swarm robotics for autonomous drone teams: Recent advances and future directions," *IEEE Robotics & Automation Magazine*, 2023.

[6] E. Nunes and M. Gini, "Multi-robot auctions for allocation of tasks with temporal constraints," in *Proc. AAAI Conf. Artificial Intelligence*, 2015.

[7] S. S. Nestinger and M. A. Demir, "A survey of consensus-based bundle algorithm variants for multi-robot task allocation," *ACM Computing Surveys*, 2024 (survey literature).

[8] A. Verma and V. Ranga, "Applications of consensus-based bundle algorithms in unmanned systems: A review," in *Proc. Int. Conf. Unmanned Aircraft Systems (ICUAS)*, 2021.

[9] S. Chopra, G. Notarstefano, M. Rice, and M. Egerstedt, "A distributed version of the hungarian method for multirobot assignment," *IEEE Transactions on Robotics*, vol. 33, no. 3, pp. 660–673, Jun. 2017.

[10] L. B. Johnson, H.-L. Choi, S. Ponda, and J. P. How, "Allowing non-submodular score functions in CBBA," in *Proc. IEEE Conf. Decision and Control (CDC)*, 2011.

[11] A. K. Whitten, H.-L. Choi, L. F. Bertuccelli, and J. P. How, "Decentralized task allocation with heterogeneous constraints," in *Proc. AIAA Guidance, Navigation, and Control Conf.*, 2010.

[12] M. C. Dimarogonas and K. H. Johansson, "Decentralized consensus-based assignment," *IEEE Transactions on Automatic Control*, 2012 (gossip consensus foundations).

[13] R. Olfati-Saber, J. A. Fax, and R. M. Murray, "Consensus and cooperation in networked multi-agent systems," *Proceedings of the IEEE*, vol. 95, no. 1, pp. 215–233, Jan. 2007.

[14] C. W. Reynolds, "Flocks, herds and schools: A distributed behavioral model," in *Proc. SIGGRAPH '87*, 1987, pp. 25–34.

[15] D. Tardioli, R. Mosteiro, E. Zema, A. González-Garrido, and A. R. Mosteo, "Grounding robotic wireless communications in realistic channel models," in *Proc. IEEE Int. Symp. Safety, Security, and Rescue Robotics (SSRR)*, 2019.

[16] J. Mott and D. Smith, "Communication architectures and challenges for multi-UUV systems in contested environments," *IEEE Communications Surveys & Tutorials*, vol. 22, no. 4, 2020.

[17] A. S. Tanenbaum and D. J. Wetherall, *Computer Networks*, 5th ed. Boston, MA: Pearson, 2011, ch. MAC protocols (TDMA/FDMA/CDMA).

[18] Y. Zhang, J. Wang, and K. Wu, "Medium access control for wireless swarm robotics: A survey of slotted approaches," *IEEE Access*, vol. 9, 2021.

[19] D. P. Bertsekas and R. G. Gallager, *Data Networks*, 2nd ed. Prentice Hall, 1992 (bounded-delay communication models).

[20] T. Kuhn, N. Lynch, and R. Oshman, "Distributed computation in dynamic networks," in *Proc. ACM Symp. Theory of Computing (STOC)*, 2010.

[21] W. H. Kruskal and W. A. Wallis, "Use of ranks in one-criterion variance analysis," *Journal of the American Statistical Association*, vol. 47, no. 260, pp. 583–621, 1952.

[22] S. S. Shapiro and M. B. Wilk, "An analysis of variance test for normality (complete samples)," *Biometrika*, vol. 52, no. 3/4, pp. 591–611, 1965.

[23] E. L. Kaplan and P. Meier, "Nonparametric estimation from incomplete observations," *Journal of the American Statistical Association*, vol. 53, no. 282, pp. 457–481, 1958.

[24] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, pp. 83–97, 1955.

[25] J. P. How et al., "Increasing mission autonomy of unmanned aerial vehicles through decentralized task allocation," *Annual Reviews in Control*, vol. 32, no. 2, 2008.